EXPERIMENT NO: 02

Date	10/07/26
Title	Sample Java Program to Check Whether the Given Number is Prime Palindrome or Not

Objective / Aim

To develop a Java program to check whether the given number is a Prime Palindrome or not using Java data types, variables, operators, control structures, methods, and console input/output.

System Requirements

Hardware
• Standard PC/Laptop
• Minimum 4 GB RAM
• Minimum 500 MB free storage

Software
• Windows / Linux / macOS
• Java Development Kit (JDK)
• Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
• Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.

2. Import the Scanner class.

3. Create a Scanner object to read input from the user.

4. Read an integer n from the user.

5. Call the isPrime(n) method to check whether the number is prime.

6. In the isPrime() method: If n <= 1, return false. If n <= 3, return true. If n is divisible by 2 or 3, return false. Check possible factors using a loop starting from 5 and incrementing by 6. If n is divisible by i or i + 2, return false. Otherwise, return true.

7. Call the isPalindrome(n) method to check whether the number is a palindrome.

8. Store the original number.

9. Reverse the number digit by digit using the % and / operators.

10. Compare the original number with the reversed number.

11. If they are equal, return true; otherwise, return false.

12. If both prime and palindrome are true, display that the number is a prime number and palindrome.

13. If only prime is true, display that the number is prime but not palindrome.

14. If only palindrome is true, display that the number is not prime but is a palindrome.

15. Otherwise, display that the number is neither prime nor palindrome.

16. Close the Scanner.

17. Stop the program.

UML Class Diagram

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■       primepalindrome        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                              ■
■ + isPrime(n: int): boolean   ■
■                              ■
■ + isPalindrome(n: int):      ■
■                    boolean     ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                ■
     static methods called by
```

```
             ■
■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■
■         Experiment2        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                            ■
■ + main(args: String[]): void ■
■                            ■
■ - pp: Scanner              ■
■ - n: int                   ■
■ - prime: boolean           ■
■ - palindrome: boolean      ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Relationship

The Experiment2 class contains the main() method and calls the static methods isPrime() and isPalindrome() of the primepalindrome class.

Source Code

Save the file as: **Experiment2.java**

```java
package recordprog;

import java.util.Scanner;

class primepalindrome {

    // method to check prime number
    static boolean isPrime(int n) {

        if (n <= 1)
            return false;

        if (n <= 3)
            return true;

        if (n % 2 == 0 || n % 3 == 0)
            return false;

        for (int i = 5; (long) i * i <= n; i += 6) {

            if (n % i == 0 || n % (i + 2) == 0)
                return false;
        }

        return true;
    }

    // method to check palindrome number
    static boolean isPalindrome(int n) {

        int original = n;
        int reversed = 0;

        while (n != 0) {

            int digit = n % 10;
            reversed = reversed * 10 + digit;
            n /= 10;
        }

        return original == reversed;
    }
}

public class Experiment2 {

    public static void main(String[] args) {

        Scanner pp = new Scanner(System.in);

        System.out.println("Enter the number");
        int n = pp.nextInt();

        boolean prime = primepalindrome.isPrime(n);
        boolean palindrome = primepalindrome.isPalindrome(n);

        if (prime && palindrome) {

            System.out.println(
                n + " is a PRIME NUMBER AND PALINDROME"
            );

        } else if (prime) {

            System.out.println(
                n + " IT IS PRIME BUT NOT PALINDROME"
            );

        } else if (palindrome) {

            System.out.println(
                n + " IS NOT PRIME BUT IT IS A PALINDROME"
            );

        } else {

            System.out.println(
                n + " IS NEITHER PRIME NOR BE PALINDROME"
            );
```

```
        }

        pp.close();
    }
}
```

Result: The Java program successfully checks whether the given number is a prime number, a palindrome, both, or neither.